

S.No	Paper Title	Publisher	Quantitative Analysis	Qualitative Analysis	Limitation
1	Smart Farming Assistant using AI for Weed Detection and Fertilizer Recommendation	IEEE Access (2026)	98.2% mAP, 97.1% accuracy	Deep Neural Network Boost with IoT	Limited regional validation
2	Smart Soil Monitoring System With Crop Health and Fertilizer Recommendation	IEEE Sensors (2024)	95.1% F1 score	IoT soil monitoring + ML	Limited crops, internet required
3	Artificial Intelligence Applications Across the Supply Chain in Smart Agriculture	MDPI Sustainability (2025)	92.5% mAP, 91.4% Precision	Review of AI in Smart Agri	Heterogeneous datasets
4	Transformer-Based Weather Prediction and Crop Recommendation	arXiv (2026)	MAE 5.31, mAP 0.87	STGNN + RAG chatbot	Limited field validation
5	Machine Learning and IoT-Driven Precision Agriculture for Crop Disease Detection	Springer Discover Environment (2026)	94.3% accuracy	ML+IoT review	High cost/interoperability
6	A Multimodal AI Framework for Explainable Crop Recommendation	MDPI Remote Sensing (2026)	3.76–4.96 F1 score	CNN+Geo+LLM	Small dataset
7	Transforming Agriculture Through AI and Robotics in Agronomy	Frontiers in Agronomy (2025)	R² up to 0.90	Comprehensive AI review	Scalability issues
8	Explainable ML-Based Crop Recommendation System	Scientific Reports (2025)	99.27% accuracy	Gradient Boosting + LIME	Limited field validation
9	AI Agriculture Assistant	Elsevier Procedia CS (2025)	99.32% accuracy	Ensemble+Chatbot+RSA	Single-language, internet
10	AgroLLM	MDPI AgriEngineering (2026)	95.2% F1, BLEU 94.5%	RAG+DKPL+LLM	Needs broader datasets